High-Level Design (HLD) — CredVault

System Overview

Field	Details
System	CredVault Credit Card Management Platform
Version	1.0
Date	April 2026

1. Introduction

CredVault is an enterprise-grade credit card management platform built using a microservices architecture.

It supports:

- User authentication
- Credit card lifecycle management
- Automated billing
- Rewards and points system
- OTP-based secure payments
- Wallet operations

This document outlines:

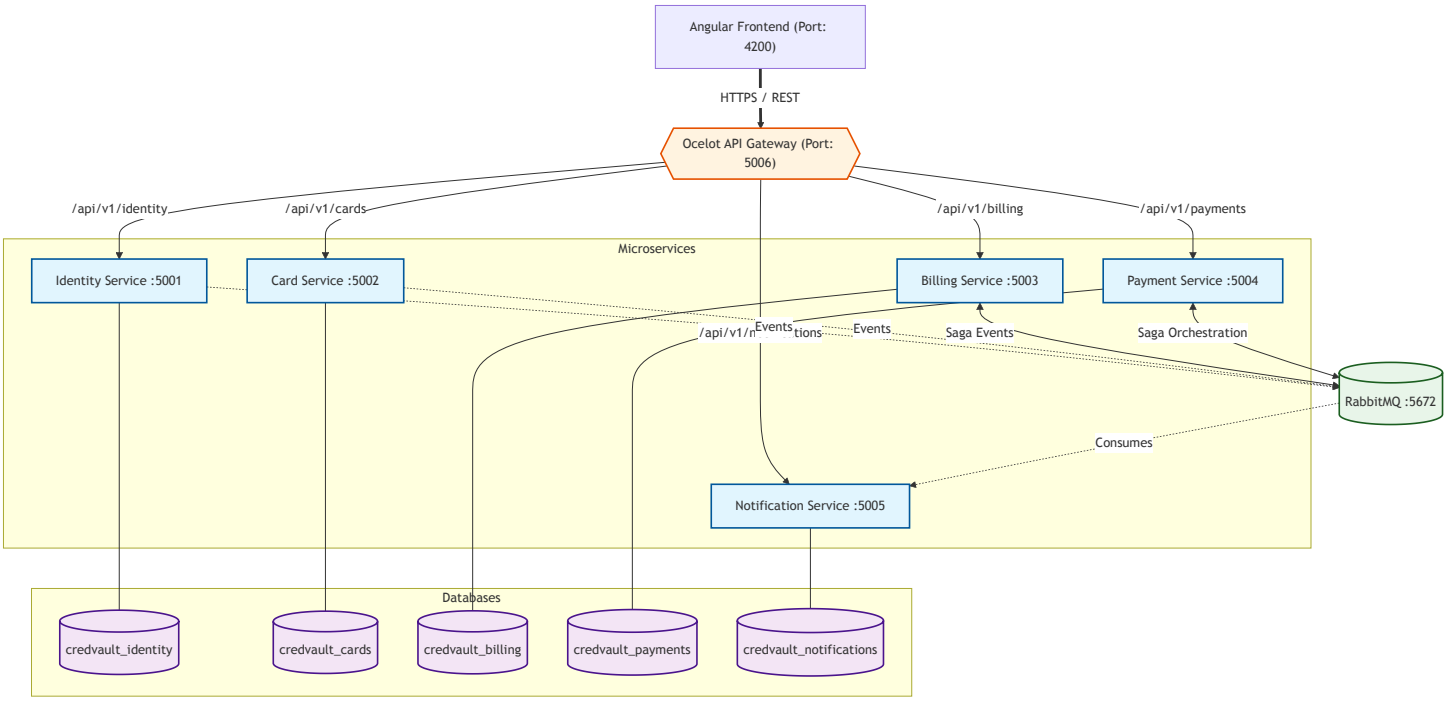
- System macro-architecture
- Service boundaries
- Communication patterns
- Technology decisions

2. System Architecture

CredVault follows a **microservices architecture** with an **API Gateway** as the single entry point. The system consists of:

- **1 Angular Frontend** (client SPA with SSR support)
- **1 Ocelot API Gateway** (request routing)
- **5 Backend Microservices** (each with its own database)
- **1 RabbitMQ Message Broker** (async inter-service communication)

2.1 Architecture Diagram



3. Service Boundaries

3.1 Responsibility Matrix

Service	Responsibility	Port	Database
Identity	User auth, JWT, OTP, profile management, RBAC	5001	`credvault_identity`

Service	Responsibility	Port	Database
Card	Card CRUD, transactions, issuer management	5002	`credvault_cards`
Billing	Bills, statements, rewards points, reward tiers	5003	`credvault_billing`
Payment	Payment flow, OTP 2FA, wallet, saga orchestration, Razorpay	5004	`credvault_payments`
Notification	Email delivery (Gmail SMTP), audit logs, notification logs	5005	`credvault_notifications`

3.2 Service Independence

Each service:

- Owns its database (no shared tables across services)
- Has no cross-service foreign key constraints
- Communicates with other services only via the Gateway (sync) or RabbitMQ (async)
- Can be deployed, scaled, and restarted independently

4. Communication Patterns

4.1 Synchronous (REST over HTTP)

Direction	Usage
Client → Gateway → Service	All user-facing API calls
The Gateway routes requests based on URL prefix:	
Route Prefix	Destination
---	---
`/api/v1/identity/`	Identity Service
`/api/v1/cards/`	Card Service

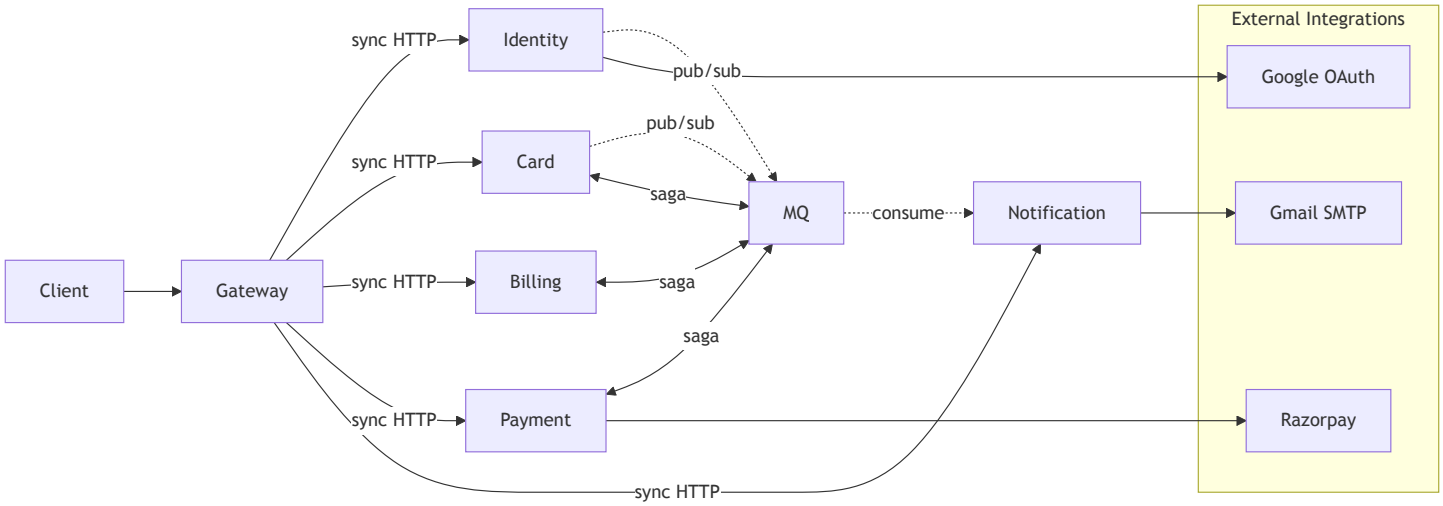
Direction	Usage
`/api/v1/issuers/`	Card Service
`/api/v1/billing/`	Billing Service
`/api/v1/payments/`	Payment Service
`/api/v1/wallets/`	Payment Service
`/api/v1/notifications/`	Notification Service

4.2 Asynchronous (RabbitMQ via MassTransit)

Pattern	Usage
Pub/Sub	Event notifications (user registered, card added, payment OTP)
Saga Orchestration	Distributed payment transactions (Payment ↔ Billing ↔ Card)
Key events:	

- `IUserRegistered` → Identity publishes, Notification consumes (sends welcome email)
- `ICardAdded` → Card publishes, Notification consumes (sends confirmation email)
- `IPaymentOtpGenerated` → Payment publishes, Notification consumes (sends OTP email)
- Saga events → Payment orchestrates with Billing and Card services

4.3 Communication Summary



5. Technology Stack

Layer	Technology
Frontend	Angular 21.2, TypeScript 5.9, Tailwind CSS 4.2, Chart.js 4.5, RxJS 7.8
Gateway	Ocelot API Gateway
Backend	.NET 8/10, ASP.NET Core Web API, C#
Messaging	RabbitMQ 3.12, MassTransit (with InMemoryOutbox)
Database	SQL Server 2022, Entity Framework Core (Code-First)
Auth	JWT Bearer tokens, Google OAuth (SSO)
Payments	Razorpay (wallet top-ups)
Email	Gmail SMTP
Logging	Serilog (structured, rolling daily logs)
Validation	FluentValidation
Containerization	Docker, Docker Compose

6. Architectural Patterns

Pattern	Implementation
Microservices	5 autonomous services, database-per-service
API Gateway	Ocelot routes all client requests by URL prefix
Clean Architecture	Domain → Application → Infrastructure → API per service
CQRS	MediatR separates Commands (writes) from Queries (reads)
Saga Pattern	MassTransit State Machine orchestrates distributed payments with compensation

Pattern	Implementation
Event-Driven	RabbitMQ pub/sub for async inter-service communication
Outbox Pattern	MassTransit `UseInMemoryOutbox()` prevents lost messages
Retry Policy	Exponential backoff (1s → 5s → 15s) on all consumers
RBAC	User and Admin roles enforced at service level
Soft Deletes	Cards use `IsDeleted` flag instead of physical deletion

7. Security Architecture

7.1 Authentication

- **JWT Bearer tokens** issued by Identity Service
- All services validate JWT using shared configuration
- Token payload: `sub`, `email`, `name`, `role`, `iss`, `aud`, `exp`
- 60-minute expiry, 30-second clock skew tolerance

7.2 Authorization

Role	Access
Public	Registration, login, OTP verification
User	Own cards, bills, payments, wallet, profile
Admin	All user management, card management, bill generation, reward tiers, logs

7.3 Data Protection

- Card numbers encrypted at rest (AES)
- Passwords hashed with BCrypt
- OTPs for email verification, password reset, and payments
- TLS for all external communication

8. External Integrations

Integration	Service	Purpose
Google OAuth	Identity Service	Passwordless SSO login
Razorpay	Payment Service	Wallet top-up payment processing
Gmail SMTP	Notification Service	Email delivery (OTP, confirmations, alerts)

9. Deployment View

9.1 Docker Compose Services

```
services:
  identity-service:       # Port 5001
  card-service:           # Port 5002
  billing-service:        # Port 5003
  payment-service:        # Port 5004
  notification-service:   # Port 5005
  gateway:                # Port 5006 (exposed to host)
  rabbitmq:               # Port 5672, Management UI: 15672
  sqlserver:              # Port 1433
```

9.2 Network Isolation

- **Public Zone**: Only Gateway (5006) is reachable from outside
- **Private Zone**: All services, databases, and RabbitMQ on isolated Docker bridge network
- Services communicate via internal DNS names (e.g., `http://identity-api:80`)

9.3 Data Persistence

- Docker volumes for SQL Server data and RabbitMQ state
- Each service runs EF Core migrations on startup (`Database.Migrate()`)
- Seed data: Card issuers (Visa, Mastercard, Rupay, Amex), default reward tiers

End of High-Level Design Document